Capstone Three: Project Ideas and Proposal

Here are three capstone project ideas, each addressing a different core topic and grounded in datasets found via Google Dataset Search.

1. Weather- and Holiday-Aware Retail Demand Forecasting (Time Series Analysis)

Problem: Retailers frequently over- or under-stock inventory because standard sales forecasts rely only on historical sales trends and ignore major external drivers of consumer demand, such as weather swings and holidays. This leads to stockouts during demand spikes (lost revenue) and overstock during lulls (waste, markdown costs). This project asks whether daily/weekly category-level sales can be forecast more accurately by combining historical sales patterns with weather conditions and holiday calendars, and which product categories are most sensitive to these external factors.

Data:

- **Retail Sales Dataset (Kaggle)** — daily sales records by date, category, region, quantity, and profit, serving as the core time series target.
- **Global Daily Climate Data / The Weather Dataset (Kaggle)** — daily temperature, precipitation, wind, and sunshine data for ~1,250 cities worldwide, used as exogenous weather features.
- **Worldwide Public Holidays 1970–2099 (Kaggle/Azure Open Datasets)** — holiday dates and types by country, used to flag holiday and pre/post-holiday demand windows.

These three datasets can be joined on date and region to build a supervised forecasting model (e.g., SARIMAX, Prophet, or gradient-boosted trees) and measure how much predictive lift the weather and holiday features add over a sales-only baseline.

2. Customer Sentiment as a Leading Indicator of Product Demand (NLP)

Problem: Businesses often learn about shifting customer sentiment only after sales have already dropped. Product reviews and social commentary frequently surface complaints, quality concerns, or growing enthusiasm well before those shifts show up in transaction data. This project investigates whether sentiment and topic trends extracted from customer review text can serve as an early warning signal for changes in product or category demand, and whether specific complaint themes (e.g., shipping delays, quality defects) correlate with measurable drops in future sales.

Data:

- A large e-commerce or product review corpus (e.g., Amazon Reviews dataset or a Yelp/Kaggle equivalent) containing review text, star ratings, timestamps, and product/category identifiers.
- A time-aligned sales or ratings-volume dataset for the same products or category, used as the outcome variable to test whether sentiment shifts precede demand shifts.

The NLP pipeline would involve sentiment scoring, topic modeling (e.g., LDA or BERTopic) to extract recurring complaint/praise themes, and time-lagged correlation or Granger causality analysis between sentiment trends and subsequent sales movement.

3. Mapping Regional Market Similarity Through Shared Demand Patterns (Network Analysis)

Problem: Retail and logistics companies often manage regional inventory and promotions independently, missing opportunities to pool stock or coordinate promotions across regions that behave similarly. This project explores whether regions can be organized into a network based on how similarly their sales respond to shared external shocks (holidays, weather events, seasonality), and whether that network reveals natural "market clusters" that could inform smarter regional inventory pooling or promotion scheduling.

Data:

- **Retail Sales Dataset (Kaggle)** — daily sales by region and category, used to compute regional demand time series.
- **Global Daily Climate Data (Kaggle)** — regional weather data, used to align demand spikes with weather events per region.
- **Worldwide Public Holidays dataset** — used to align regional demand with shared or region-specific holidays.

Regions would become nodes in a network, with edge weights derived from the correlation or co-movement of their sales responses to the same weather/holiday events. Community detection algorithms (e.g., Louvain, Girvan-Newman) would then be applied to identify clusters of regions with similar demand behavior, which could be visualized and interpreted for business implications like shared warehousing or synchronized promotions.

As a Final Capstone Project, I would like to implement a use case that is built around ***time series analysis***.

This Project uses the three datasets I pulled from Google Dataset Search and the reasoning behind combining them.

Capstone Project: Weather- and Holiday-Aware Retail Demand Forecasting

The problem. Retailers routinely over- or under-stock inventory because standard sales forecasts ignore two of the biggest external drivers of consumer behavior: weather and holidays. A grocery chain that doesn't anticipate a heatwave-driven spike in beverage sales, or a holiday-driven spike in gift categories, ends up with stockouts (lost revenue) or overstock (waste, markdowns, storage cost). The capstone asks: **can we build a forecasting model that predicts daily/weekly sales by product category more accurately by incorporating weather conditions and holiday calendars alongside historical sales patterns?** This also lets you explore secondary questions — which categories are most weather-sensitive (e.g., ice cream vs. office supplies), and how much lead time before a holiday sales start ramping up.

The three datasets

1. **Retail Sales Dataset (Kaggle)** — daily retail sales data with sales performance, product category trends, and regional profitability, including date, category, sales revenue, quantity sold, profit, and region. This is the core time series: your target variable (sales) at daily granularity, segmented by category and region. kaggle.com/datasets/abbas829/retail-sales-dataset
2. **The Weather Dataset / Global Daily Climate Data (Kaggle)** — daily weather readings from ~1,250 cities worldwide, including average/min/max temperature, precipitation, snow depth, wind speed, sea-level pressure, and sunshine duration. This becomes your exogenous weather feature set, joined to the retail data on date (and region/city if you match a specific market). kaggle.com/datasets/guillemservera/global-daily-climate-data
3. **Worldwide Public Holidays 1970–2099 (Kaggle / Azure Open Datasets)** — holiday data spanning 38 countries or regions from 1970 to 2099, with each row giving the date, country/region, and whether it's a paid day off. This lets you flag holiday and pre/post-holiday windows as features. kaggle.com/datasets/fridrichmrtn/public-holidays

How they combine

- Join retail sales (dataset 1) to weather (dataset 2) on **date + region**, and to holidays (dataset 3) on **date + country/region**.
- Engineer features: temperature deviation from seasonal norm, precipitation flags, "days to/from nearest holiday," holiday type, and rolling averages of sales.
- Build baseline models (naive seasonal, ARIMA/SARIMA) then compare against models with the weather/holiday regressors (SARIMAX, Prophet with regressors, or gradient-boosted trees like XGBoost/LightGBM treating it as a supervised forecasting problem).
- Evaluate with MAPE/RMSE, and specifically measure the **lift from adding weather and holiday features** over a sales-only baseline — that's your core research contribution.